

Session 5:

Git & Version Control Why version control is needed. Git concepts: Repository, Commit, Branch, Merge, Push, Pull. GitHub overview: Forks, Pull Requests, Collaboration. Trainer Demo: Initialize a local Git repo (git init, git add ., git commit) Push to GitHub. AI Integration: Use GitHub Copilot Chat to explain git errors in real-time. Depth: Students should confidently push their first small project to GitHub.

- 1) Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

Initialize a repository

```
</> Bash  
git init
```

Commit with a message

```
</> Bash  
git commit -m "Initial commit"
```

Push to remote repository

```
</> Bash  
git push origin main
```

Pull latest changes

```
</> Bash  
git pull origin main
```

2) Difference between git fetch and git pull (2 lines)

- git fetch downloads changes from the remote repository but does not merge them into your local branch.
- git pull downloads changes and automatically merges them into your current branch.

3) Create a GitHub repository and push a file

Example commands:

`</> Bash`

```
echo "Hello GitHub" > sample.txt
git init
git add sample.txt
git commit -m "Add sample file"
git branch -M main
git remote add origin https://github.com/your-username/my-repo.git
git push -u origin main/...
```

Repo Link:

<https://github.com/your-username/my-repo>

4) Difference between a Fork and a Pull Request (2 lines)

- A **Fork** is a personal copy of someone else's GitHub repository in your account.
- A **Pull Request (PR)** is a request to merge your changes from a fork or branch into another repository.

5) Create branch feature-test, commit, and merge into main

</> Bash

```
# Create and switch to new branch
git checkout -b feature-test
```

```
# Make changes to a file
echo "New feature" >> sample.txt
```

```
# Stage and commit changes
git add .
git commit -m "Added feature-test changes"
```

```
# Switch back to main branch
git checkout main
```

```
# Merge feature-test into main
git merge feature-test
```

Commands used:

</> Bash

```
git checkout -b feature-test
```

```
git add .
```

```
git commit -m "Added feature-test changes"
```

```
git checkout main
```

```
git merge feature-test
```